Recognition in Label and Discrimination in Feature: A Hierarchically Designed Lightweight Method for Limited Data in SAR ATR

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***Abstract*— Synthetic aperture radar (SAR) automatic target recognition (ATR) is an essential field in SAR application. However, a sufficient number of labeled training SAR images for each target type plays a crucial role in existing SAR ATR methods, while the acquisition and annotation of SAR images are difficult and time consuming in practice. Therefore, the recognition under the limited labeled training SAR images is the basic and crucial problem in SAR application. In this article, we propose a novel hierarchically designed lightweight method (HDLM) by recognition in label and discrimination in feature to address the problem of limited data in SAR ATR. The proposed method is hierarchically designed from top to bottom. In the top phase, the framework is constructed by dual loss to force the deep model to optimize by label recognition and feature discrimination, which is noted as recognition in label and discrimination in feature. In the middle phase, the architecture of the network is built up using a novel lightweight (LW) extractor and multilevel cross fusion to boost the amount and diversity of the features for the framework. In the bottom phase, two modules, coordinate attention, and depth-wise separable convolution modules are employed to enhance the feature quality and density with fewer parameters for the phases above. The experimental results on moving and stationary target acquisition and recognition (MSTAR) and OpenSARship showed that the proposed HDLM performs better than the existing methods under the limited training samples.**

***Index Terms*— Automatic target recognition (ATR), discrimination in feature, hierarchically designed lightweight method (HDLM), limited data, recognition in label, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an important microwave remote sensing system because of its advantages of collecting all-weather and day-and-night high-resolution images [1]. In particular, SAR automatic target recognition (ATR) appears to be one of the most crucial and demanding challenges in SAR applications [2]. For the past decades, scholars have proposed numerous novel classification algorithms and systems. These different approaches can be grouped into three dominant paradigms: template-based, model-based, and deep learning [3], [4], [5], [6]. The template-based classification approach can be regarded as a pattern recognition method that relies entirely on the template as a representation of the target [7], [8]. The model-based classification method focuses mainly on the features that represent the physical structure of the target [9], [10].

Recently, when the achievements of deep learning applied to the field of signal and image processing proved its superior performance [11], [12], many novel studies on SAR ATR have been proposed with outstanding results [13], [14], [15], [16]. Although the current SAR ATR deep learning methods have achieved great performance, they require hundreds and thousands of SAR image samples to train the deep network to avoid overfitting and acquire great generalization performance of recognition under various imaging scenarios.

However, the differences in the characteristics of SAR images under various imaging scenarios inevitably raise the problem of insufficient training data in SAR ATR. For the success of the classification, it is usually essential to gather a huge number of SAR images under various imaging scenarios, such as target configuration, azimuth, and depression. In the actual situation, gathering such sufficient SAR images is still a challenge for the following reasons.

1) The acquisition of SAR images is more consuming than that of optical images.

2) The manual annotation of SAR images is very time consuming and in high professional demand. The lack of sufficient training SAR images can greatly hinder the performance and application of SAR ATR methods. At the same time, the problem of insufficient SAR images is basic and crucial in SAR civil application. More specifically, different shapes or sizes of building or vehicles will lead to the problem of insufficient SAR images, let alone the various platforms and imaging conditions.

Several attempts have been recently made to tackle this problem of SAR ATR. These methods can be divided into two types based on the problem formulation: limited SAR images and few-shot learning (FSL). The problem of limited SAR images has training and testing datasets. The sample numbers

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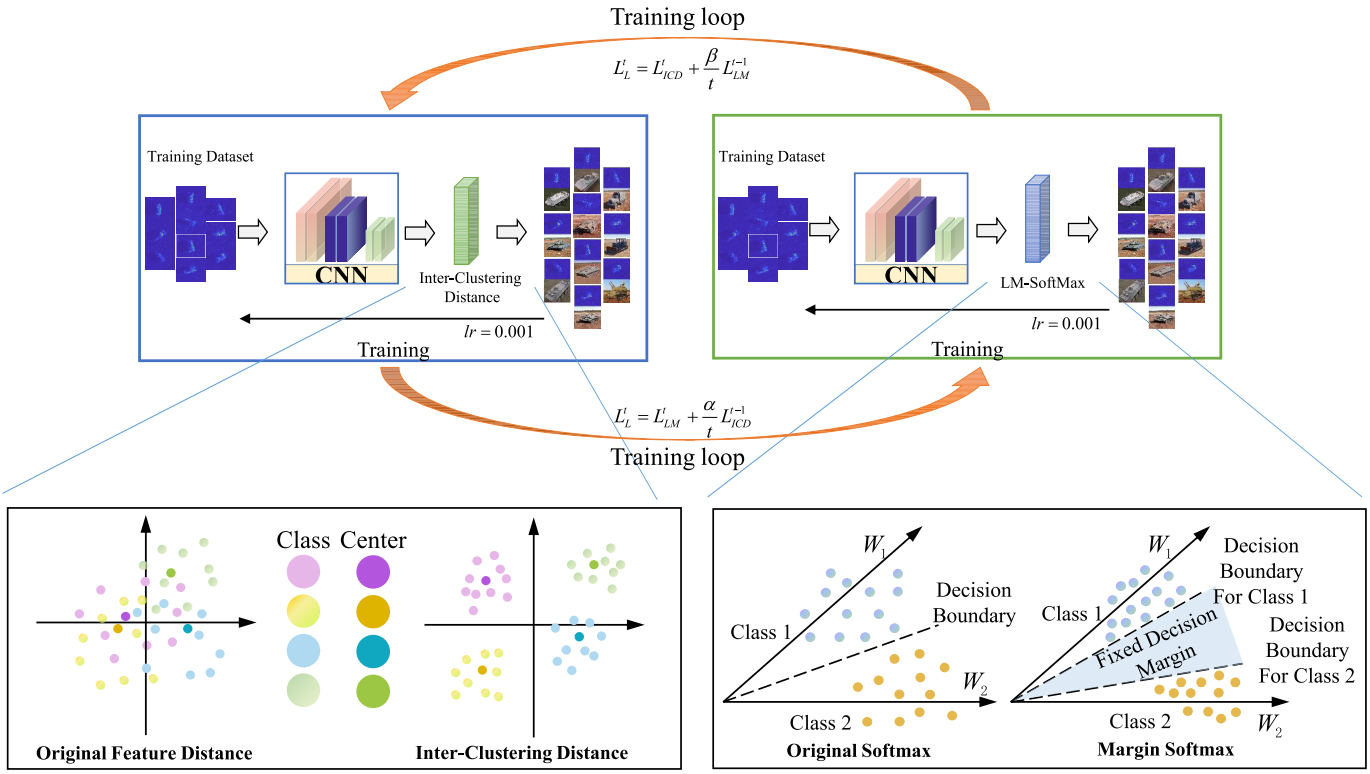


Fig. 1. (Top Phase) Framework of proposed method. A CNN is trained with losses of intraclustering distance and LM-SoftMax. Process of training loop is iterative which is named as the progressive harmony stage, one time for the loss of intraclustering distance, one time for the loss of LM-SoftMax.

for each class type in the training dataset are limited. The FSL problem also has a training dataset with a limited number of samples for each class type, but there is a support dataset with a sufficient number of samples for class types in the testing dataset. The class types are disjoint between the training and testing dataset.

In methodology, these methods can be divided into data augmentation and deep model-based. Data augmentation methods use prior knowledge to augment data to enhance the amount and enrich the quality of training samples [17], [18], [19]. For example, Wang et al. [20] designed a semisupervised learning framework, including a self-consistent augmentation rule, mixup-based labeled and unlabeled mixture, and weighted loss, to utilize unlabeled data during training. Cao et al. [19] proposed an alternative SAR image generated method based on label-directed GANs (LDGANs), which can control the generation of SAR image data of a certain class. These methods need sufficient SAR images to construct the manifold structure similar to few labeled training samples, which do not always exist.

Deep model-based methods mainly depend on architecture design for rapid generalization of FSL tasks. For example, Fu et al. [21] employed meta-learning in the few-shot SAR target classification problem and utilized three transfer learning methods to alleviate the difficulties of meta-learning with fewer training classes. Wang et al. [22] proposed a hybrid inference network, including embedding network and hybrid inference strategy stages, which obtained good performance of three-target classification. Zhou et al. [23] proposed a large-margin SoftMax (LM-SoftMax) batch-normalization convolutional neural network (CNN) to obtain good generalization performance and high convergence speed. These methods need sufficient label SAR images under quite similar imaging conditions to develop a helpful inductive bias for high recognition performance.

However, these methods still face the two challenges above: difficult acquisition and time-consuming annotation. To overcome these challenges in SAR ATR, our key motivation is to promote the effectiveness of the extracted features by recognition in label and discrimination in features. To be more specific, unlike the existing SAR ATR that uses limited data only based on label recognition, our method not only recognizes in label, but also exploits the discrimination in features by enhancing the intraclass feature clustering and interclass feature distancing called intraclustering distance, as shown in Fig. 1. Additionally, to deal with the overfitting of deep learning methods in the limited data or FSL, our method focuses on reducing the number of parameters and simplifying hypotheses using novel lightweight (LW) modules.

To tackle with the problem of limited training data, this article proposes a novel hierarchically-designed lightweight method (HDLM) by recognition in label and discrimination in feature for SAR ATR. The main contributions of this article are summarized as follows.

1) In the top phase, we propose a novel framework constructed by dual loss to construct and optimize the embedding space to learn for label recognition and feature discrimination

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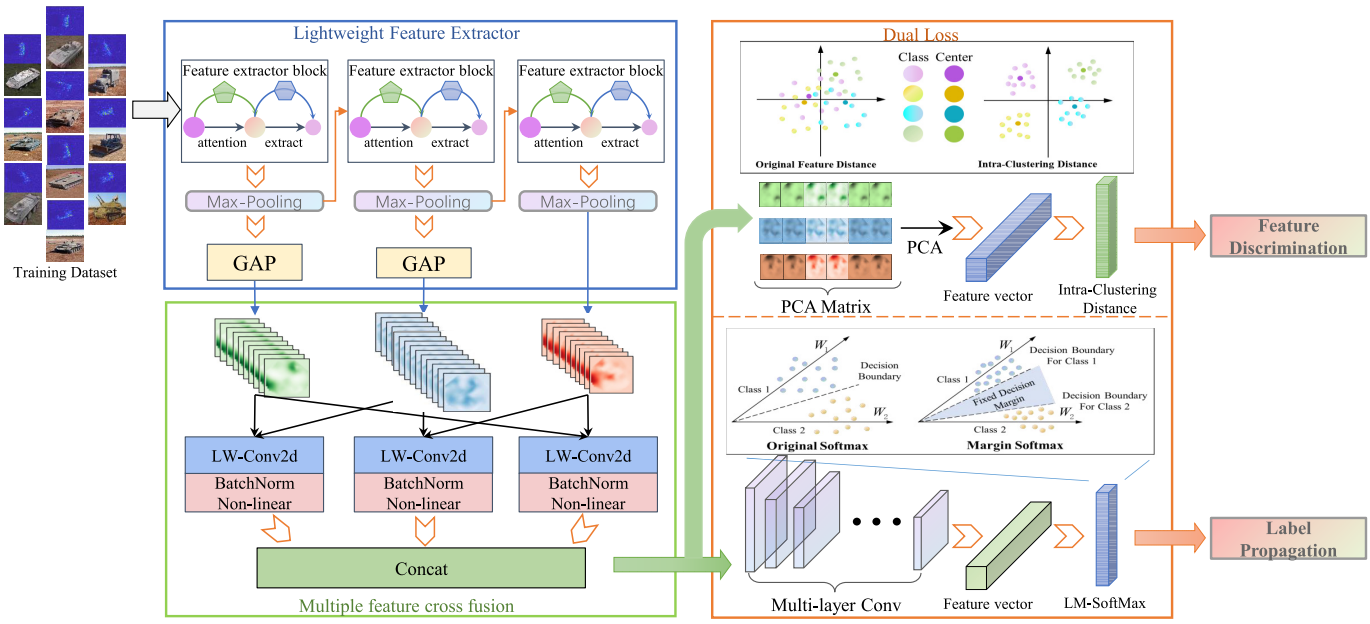


Fig. 2. (Middle Phase) Architecture of proposed HDLM. Green pentagon in the feature extractor block presents the coordinate attention module, and blue hexagon in feature extractor block is lightweight convolutional layer. GAP means the global average pooling layer [24], Nonlinear means nonlinear function, LW-Con2d means lightweight 2-D convolutional layer.

(Fig. 1). The dual loss can force the model to enhance the effectiveness of the extracted feature by both interclustering distance and LM-SoftMax for limited training data.

2) In the middle phase, we proposed a novel architecture (Fig. 2), including a lightweight extractor (blue box in Fig. 2) and multilevel cross fusion (green box in Fig. 2), to extract and fuse multilevel effective features with fewer parameter numbers. The architecture is designed to provide more sufficient and diverse features on different target scale with fewer parameter numbers (Fig. 2) for limited training data.

3) As a result, the proposed HDLM achieved superior recognition performance under limited data of moving and stationary target acquisition and recognition (MSTAR) and OpenSARship, when the number of training samples ranges from 150 in each class to 5 in each class. We also validate the capability of the proposed HDLM under the recognition problems of FSL. The ablation experiments indicated the effectiveness of the proposed HDLM in recognition and feature discrimination.

Last but not least, in the bottom phase, two specific modules are employed. Depth-wise separable convolution is a lightweight convolution to avoid the overfitting under limited training data. Coordinate attention focus on important target features to enhance the feature effectiveness for limited training data.

The remainder of this article is organized as follows. An overview of the HDLM is presented in Section II. Section III validates the effectiveness of the proposed method through experiments. Finally, Section IV presents the conclusion.

II. Proposed Method

In this section, we discuss the theoretical challenge of limited data briefly and summarize the difficulty of deep learning with limited data to explain the motivation of the proposed method. Then, the dual loss-based framework is described in detail to address the corresponding difficulty. Finally, the architecture in the middle phase and modules in the bottom phase matching the framework in the top phase are presented.

A. Theoretical Challenge of Limited Data in SAR ATR

As described in [25], the well-known generalization of the model on the testing dataset satisfies the equation of the training sample number and the model complexity as

$$P(\text{test error} \le \text{training error} + \text{mcp}) = 1 - \eta \tag{1}$$

where mcp is the acronyms of model complexity penalty, $\eta \in (0, 1]$. The calculation of mcp is shown as

$$\text{mcp} = \sqrt{\frac{h(\log(2N/h) + 1) - \log(\eta/4)}{N}} \tag{2}$$

where N and h denote training sample number and model complexity, respectively. From these equations above, it is clear that when training error and model complexity penalty are small, the model in the test has a good chance to perform superiorly. In general, there are several factors that determine the model complexity, including model framework, model size, etc [26].

However, in practical scene, the model needs to go deeper or wider to extract more effective features to achieve a small training error. It means more model complexity h, but the model complexity penalty increases together, which hinders the performance of recognition under few training samples in SAR ATR. Therefore, under few training samples, promoting the effectiveness of the extracted features should be considered instead of increasing the sample number N to improve the generalization in the testing dataset.

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Meanwhile, under few training samples, when the network goes deeper or wider, and the parameters of the network are heavily increased, the network performs better in training. However, the performance in testing inevitably goes worse with overfitting under few training samples. Therefore, to solve the FSL problem in SAR ATR, it is necessary to reduce the model complexity.

Thus, we propose a novel method called HDLM for SAR ATR, with three main phases: 1) top phase: dual loss-based training in the framework; 2) middle phase: lightweight extractor and multilevel cross fusion in architecture; and 3) bottom phase: coordinate attention and depth-wise separable convolution in module.

The three phases construct the whole method from top to bottom. The top phases (Section II-B) make the whole method to enhance the effectiveness of the features by recognition in label and discrimination in feature for the problem of limited training data. The middle phase (Section II-C) provides more diverse features on different image scale for the optimization of recognition in label and discrimination in feature. The bottom phase (Section II-D) is the basic modules to construct the whole architectures with more effective feature and few parameter numbers. The details of the proposed method are presented as follows.

B. Framework in Top Phase: Dual Loss-Based Training Loop

As discussed above, to address the problem of the limited training samples, the effectiveness of the extracted features should be promoted to improve the generalization in the testing dataset. Therefore, we proposed the dual loss with two parts: LM-SoftMax and interclustering distance. The proposed dual loss promotes the recognition ratio in label recognition and feature discrimination in the intermediate embedding space. The corresponding training process is proposed to achieve the progressive harmony of the two parts. The details of the corresponding training process and dual loss are as follows.

1) Training Loop: As shown in Fig. 1, the framework of the proposed method is training the model with progressive harmony.

In the training stage, the few sample training dataset is input to the proposed network, the feature maps go through the CNN consisting of the coordinate attention and lightweight convolutional layers, and the parameters of the entire network are learnable. At this time, the loss of interclustering distance is employed only to construct the embedding space and train the embedding network. The LM-SoftMax is employed only to promote the classifier and train the last few convolutional layers. The two parts of loss are alternated once in each step of training.

Then, in the progressive harmony stage [27], the two parts of loss are combined to train the entire network by setting a training step t to control the combination progressively. At the tth step, set the loss of interclustering distance L_{ICD}^{t} and the loss of LM-SoftMax L_{LM}^{t}. When the LM-SoftMax dominates the total loss, it is scaled as

$$L_L^t = L_{\mathrm{LM}}^t + \frac{\alpha}{t} L_{\mathrm{ICD}}^{t-1} \tag{3}$$

where α denotes the parameters that adjust the ratio of interclustering distance at the $(t-1)$th step L_{ICD}^{t-1}. At the initial step $t = 1$, the losses are set as $L_L^1 = L_{\mathrm{LM}}^1$.

When the interclustering distance dominates the total loss, at the tth step, it is scaled as

$$L_L^t = L_{\mathrm{ICD}}^t + \frac{\beta}{t} L_{\mathrm{LM}}^{t-1} \tag{4}$$

where β denotes the parameters that adjust the ratio of LM-SoftMax at the $(t-1)$th step L_{LM}^{t-1}.

After sufficient training epochs, the testing dataset is inputted to the network to measure the effectiveness of the proposed network.

Through this progressive harmony stage, the dual loss can lead to more effective optimization separately for embedding network and classifier. Then, during training, the integration of the losses of interclustering distance and LM-SoftMax can help the entire model optimize to achieve better collaboration and higher accuracy of the recognition.

2) Interclustering Distance for Feature Discrimination: The interclustering distance aims to exploit the discrimination in feature and enhance the intraclass feature clustering and interclass feature distancing. Therefore, we introduce a concept, mutual features, which are the shared features among the SAR images of one target type. As shown in Fig. 2 (the upper half part in the orange block), consider a classification task with N samples and K classes, at first the mutual features m_k in the kth type of target are extracted from the flatten feature maps using principal component analysis (PCA). For all samples, $x_1, x_2, \ldots, x_i$, in the kth type of target, the Euclidean distance is employed to calculate the distance between one sample x_i and mutual features m_k as follows:

$$d(\zeta(x_i), m_k) = \|\zeta(x_i) - m_k\|^2 \tag{5}$$

where $\|\cdot\|^2$ denotes L2-norm and $\zeta(\cdot)$ denotes the output of the flatten feature maps after intermediate embedding and PCA.

The Euclidean distance between mutual features of different target types, m_k and m_l, is calculated by

$$d'(m_k, m_l) = -\|m_k - m_l\|^2 \tag{6}$$

where m_l is the mutual features of the lth target type. Then the interclustering distance for feature discrimination is calculated as

$$L_{\mathrm{ICD}} = \frac{1}{N \times K} \sum_{k=1}^{K} \sum_{i=1}^{N} d\left(\zeta\left(x_i^k\right), m_k\right) + \frac{1}{K \times (K-1)} \sum_{i \neq k} \sum_{k=1}^{K} d'(m_i, m_k). \tag{7}$$

The first summation of (7) forces each sample of the kth target cluster near the mutual features m_k, which is actually intraclass clustering. The second summation of (7) aims to move mutual features of different target types far away, which is interclass feature distancing.

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3) LM-SoftMax for Label Recognition: Besides the feature discrimination, as shown in Fig. 1 right bottom, the LM-SoftMax can greatly improve intraclass compactness and interclass separability in the label recognition than normal SoftMax. Consider a classification task with N samples and K classes [23], $f_{x_i} = \{f_1, f_2, \ldots, f_K\}$ is the output vector of the network before normal SoftMax with the input x_i, the probability of the sample x_i classified to class y_i is

$$p_i(x_i) = \frac{e^{f_i}}{\sum_{k=1}^{K} e^{f_k}} \tag{8}$$

where $k \in [1, K]$ is the label and f_i is the ith number in f_{x_i}. Then, the loss equation of SoftMax is list as

$$L_i(x_i) = -\log\left(\frac{e^{f_i}}{\sum_k e^{f_k}}\right). \tag{9}$$

In other words, if the sample x_i is classified correctly, (9) means for $k \in [1, K]$

$$f_i = W_{y_i}^T \partial(x_i) > f_k = W_k^T \partial(x_i) \tag{10}$$

where $\partial(x_i)$ denotes the intermediate embedding of the input x_i, W_{y_i} and W_k mean the weight vectors of the multilayer convolutions.

The LM-SoftMax requires the $\|W_{y_i}\|\|\partial(x_i)\|\cos(m\theta_{y_i}) > \|W_k\|\|\partial(x_i)\|\cos(\theta_k)$. Here θ_{y_i} is the angle between W_{y_i} and x, θ_k is the angle between W_k and x, m is a positive integer. When m increases, the classification margin becomes larger and the learning objective becomes harder [28]. The LM-SoftMax has more strict decision inequality,

$$\begin{aligned}\|W_{y_i}\|\|\partial(x_i)\|\cos\left(\theta_{y_i}\right) &> \|W_{y_i}\|\|\partial(x_i)\|\psi\left(\theta_{y_i}\right) \\ &> \|W_k\|\|\partial(x_i)\|\cos(\theta_k)\end{aligned} \tag{11}$$

where the operator $\psi(\theta)$ satisfies $\psi(\theta) = (-1)^q\cos(m\theta) - 2q$, where $\theta \in [(q\pi/m), ((q+1)\pi/m)]$, $q \in [0, m-1]$. Then, different from SoftMax in (8), the equation of LM-SoftMax is present as

$$L(y_k) = \frac{e^{\|W_{y_i}\|\|\partial(x_i)\|\psi(\theta_k)}}{e^{\|W_{y_i}\|\|\partial(x_i)\|\psi(\theta_k)} + \sum_{k\neq y_i} e^{\|W_k\|\|\partial(x_i)\|\cos(\theta_k)}}. \tag{12}$$

Finally, the loss of LM-SoftMax is presented as

$$L_{\mathrm{LM}}(x_i, y_i) = -\sum_{i=1}^{K} y_i \log(L(y_k)). \tag{13}$$

Through dual loss, the proposed HDLM framework can promote the recognition ratio in label recognition and the feature discrimination in the intermediate embedding space. To guarantee the effectiveness of the proposed HDLM framework, it is necessary to carefully construct the architecture of the proposed HDLM.

C. Architecture in Middle Phase: Lightweight Extractor and Multilevel Cross Fusion

From shadow layers to deep layers in the architecture of CNNs, the extracted features in different levels of the network focus on different scales of the target features. For example, the shadow feature maps of CNNs focus more on the local features. The deep feature maps prefer more global features, such as the whole structure of the target. The recognition is sensitive to the effectiveness and density of features input to the classifier because the network depth and width are limited by the limited training samples. Therefore, to guarantee the effectiveness of interclustering distance and LM-SoftMax, we focus on improving the effectiveness of features extracted in each level of CNNs and enhancing the feature density for the classifier by fusing multilevel features.

The basic architecture of the HDLM for SAR ATR is proposed, as shown in Fig. 2. The architecture consists of two innovations: a lightweight extractor (blue block) and multilevel cross fusion (green block). The lightweight extractor forces the extracted features to focus on important target features by ignoring unnecessary features. It promotes the effectiveness of features in each level of the network and decreases the number of parameters of the entire network. Multilevel cross fusion combines the feature maps of different scales. It provides more feature density for label recognition and more feature scales for feature discrimination. The two innovations are described as follows.

As shown in the blue block in Fig. 2, a lightweight extractor is a bottom-up pathway with more feature effectiveness and fewer parameters than the normal convolutional layer. The structure of the lightweight extractor is constructed by stacking depth-wise separable convolutions. It also has the tight junction property of dense block [29] by employing a skip line from input to output of the depth-wise separable convolution. In each feature extractor block, the green pentagon indicates that the feature maps go through the coordinate attention (presented in Subsection D). The blue hexagon indicates that the feature maps go through the depth-wise separable convolutional layer (presented in Subsection D).

Multilevel cross fusion is proposed by the motivation of the following facts. For the features extracted by CNNs, the key features of different types of targets are distributed in different levels of these features. A simple stack of different level features, limited by few training samples, only increases the parameter and degree of overfitting without promoting the performance of recognition. Multilevel cross fusion adaptively fuses the multilevel features in pairs to enhance the efficient feature input to the classifier and reduce the number of parameters of the subsequent network. This can significantly increase the feature density for the classifier of label recognition and further benefit for limited training samples.

As shown in the green block in Fig. 2, multilevel cross fusion consists of three main parts. First, the features from different levels are scaled to the same size for preliminary fusion using the skip line and global average pooling. Then, according to different hierarchies of the features, these multilevel features are adaptively fused in pairs through a lightweight convolutional layer and batch normalization. Finally, the fused features are concatenated to provide more feature density for the recognition.

In the structures of the lightweight extractor and multilevel cross fusion, coordinate attention and depth-wise separable convolution modules are employed to improve the effectiveness of features extracted in each level of CNNs and enhance

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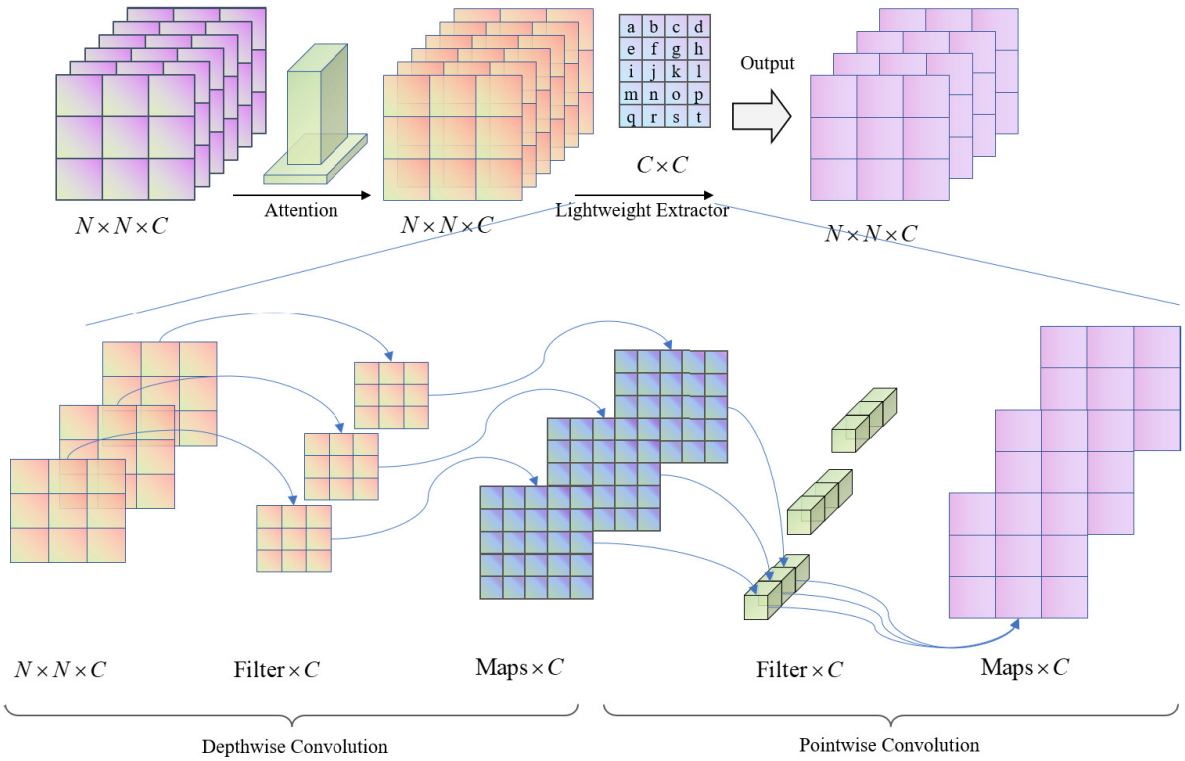


Fig. 3. (Bottom Phase) Depth-wise separable convolution. The depth-wise separable convolution is based on that the correlation between channels and spatial in the convolution layer can be decoupled. Depth-wise separable convolution has two steps, depth-wise convolution and pointwise convolution.

the feature density of the input of the classifier by fusing multilevel features. The details of coordinate attention and depth-wise separable convolution are described as follows.

D. Module in Bottom Phase: Coordinate Attention and Depth-Wise Separable Convolution

Limited by few training samples, the depth and width of the network face the tradeoff between feature effectiveness and parameter number. Because the feature effectiveness is related to the depth and width of the network, if the network goes deeper or wider, the parameter will face rapid growth, leading to overfitting under few training samples. Promoting feature effectiveness in lightweight structure seems like a paradox. Therefore, we employed coordinate attention [30] to focus on important target features while suppressing unnecessary features, and depth-wise separable convolution [31] with much fewer parameters to tackle the challenge of the limited data for SAR ATR.

The coordinate attention encodes channel and positional information by two steps [30]. First, the coordinate attention factorizes the global pooling into a pair of 1-D feature encoding operations. Then it concatenates the features along with the two spatial directions and sends them to a shared 1×1 convolutional transformation function with batch normalization and nonlinear activation function, and two separate 1×1 convolutional transformations with the sigmoid function to more accurately locate and recognize the objects of interest.

The depth-wise separable convolution can work with the coordinate attention to further promote the effectiveness of the extracted feature for better performance of the limited-data SAR ATR. There are two steps in depth-wise separable convolution: depth-wise and pointwise convolutions, as shown in Fig. 3.

The depth-wise separable convolution can extract features with much fewer parameters than the normal convolutional layer, which is beneficial for the limited-data SAR ATR. The complexity in our method is calculated to show that the computational complexity of the network is reduced. For a standard convolution layer, the time complexity is in the order of $O(N_1 \cdot N_2 \cdot W_1 \cdot W_2 \cdot C_{\text{in}} \cdot C_{\text{out}})$. Here, the size of the output feature map is $N_1 \times N_2$; the size of the convolution kernel is $W_1 \times W_2$; C_{in} denotes the number of channels in each convolution kernel; C_{out} denotes the number of kernels in this layer. Each input channel will only be processed by one corresponding convolution kernel, reducing the redundancy of the model. Meanwhile, the time complexity of depth-wise separable convolution is in the order of $O(N_1 \cdot N_2 \cdot W_1 \cdot W_2 \cdot C_{\text{in}} + N_1 \cdot N_2 \cdot C_{\text{in}} \cdot C_{\text{out}})$. Therefore, the computational complexity is reduced in our method.

For the reduced parameter set, there is a specific example as follows. Suppose the input channel of one layer is eight and the output channel is 16, and the convolutional kernel is 3×3, then the parameters of the normal way are calculated as $(8 \times 3 \times 3 + 1) \times 16 = 1168$, where 1 is the parameter of the bias. The parameters of depth-wise separable convolution are calculated as $8 \times 3 \times 3 + (8 \times 1 \times 1 + 1) \times 16 = 216$. Through the depth-wise separable convolution, the decreased parameters are $(1168 - 216/1168) = (952/1168) = 81.50\%$.

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Therefore, our method has much less parameter numbers than normal deep learning models.

The proposed HDLM first employed the dual loss and training loop to construct and optimize the embedding space toward the capability of learning for label recognition and feature discrimination. Then, the novel architecture of lightweight extractor and multilevel cross fusion is proposed to provide more effective and comprehensive features for the dual loss by reducing the number of parameters and simplifying the hypotheses using the novel lightweight modules, coordinate attention, and depth-wise separable convolution. Furthermore, we conducted experiments with limited data to validate the effectiveness and practicability of the proposed method.

III. Experiments and Results

In this section, we first introduce the MSTAR and OpenSARShip datasets, the benchmark datasets for SAR ATR. Then, we run the ablation experiments under limited training data to validate our innovations to enhance the extracted features and improve the recognition performance. Finally, we run two types of experiments, the experiments of the limited training samples (ELTSs) and the experiments of FSL (EFSL). To validate the effectiveness of the proposed method under different degrees of limited training data, all experiments are conducted under two benchmark datasets where the training samples in each class are distributed in a wide range.

A. Dataset and Preprocessing

The experiment dataset used to evaluate the proposed HDLM contains MSTAR and OpenSARShip datasets.

MSTAR was released by the Defense Advanced Research Projects Agency and the Air Force Research Laboratory and collected using the Sandia National Laboratory STARLOS sensor platform. As a benchmark dataset for SAR ATR performance evaluation, the dataset has a large number of SAR images that contain various types of military vehicles and clutter images. Ten different classes of ground targets containing tanks, rocket launchers, armored personnel carriers, air defense units, and bulldozers, are obtained as 1-ft resolution X-band SAR images with full aspect coverage (in the range of 0° to 360°). They are captured under different operating conditions, such as different aspect angles, depression angles, and serial numbers. These ten different classes of ground targets have different serial numbers under depression angles. Fig. 4 shows the SAR and corresponding optical images of ten targets at similar aspect angles.

The OpenSARShip dataset is constructed to develop sophisticated ship detection and classification algorithms under high interference. It is collected from 41 Sentinel-1 images with various environmental conditions, providing 11 346 SAR ship chips integrated with automatic identification system (AIS) messages covering 17 types. The dataset adopted in this experiment is a ground range detected product under Sentinel-1 interferometric wide (IW) mode with 2.0 m × 1.5 m resolution and 10 m × 10 m pixel size in azimuth and range directions. The length of the ships ranges from 92 to 399 m, while the width ranges from 6 to 65 m. The greatest characteristic of this dataset is that the labeling of ships is based on AIS information so that the data labeling has high reliability [32]. In our experiment, we decided to use the vertical-vertical (VV) and vertical-horizontal (VH) data, meaning that these two data types can be used in the training, validation, and test process. Fig. 5 shows that the SAR images of the three-class target.



Fig. 4. SAR images and corresponding optical images of targets.

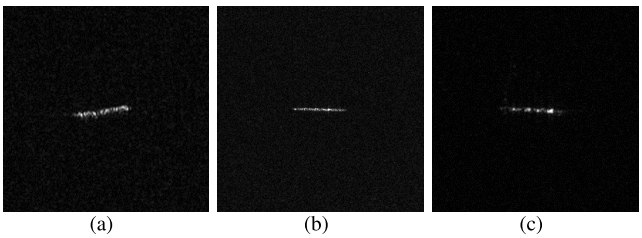


Fig. 5. Images of three-class targets in the OpenSARShip dataset. (a) Bulk carrier. (b) Container ship. (c) Tanker.

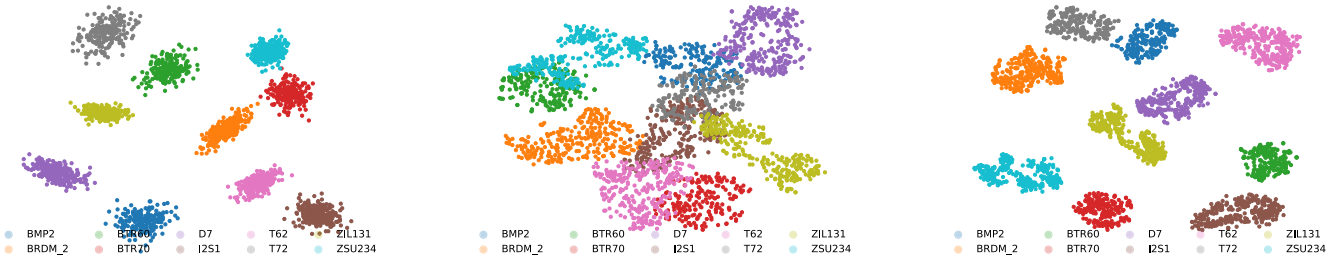
The configurations of the training and the network are present as follows. The input SAR images are of size 88 × 88. For each lightweight attention feature extractor, the convolution kernels of two stages of the depth-wise separable convolution are set as 3 × 3 and 1 × 1. For the multilevel feature cross fusion, we employ the concat to achieve the feature fusion. For each max-pooling layer, the size of the stride is set as 2 × 2. For the multilayer Conv in Fig. 2, one convolution layer is followed by one batchnorm layer and ReLu. The batch size is set as 128, and the learning rate is initialized as 0.001 and reduced with the 0.75 ratios for every ten epochs. There are also five epochs to warm up for training.

In this article, to avoid confusion, the ELTS mean that the training samples of every target type are limited, such as 10 or 20 for each class type. The EFSL refer to that the sample numbers of target types are limited in the training dataset, whereas the training samples in the support dataset are sufficient in FSL. The class types are disjoint between the training and support datasets.

B. Method Soundness Verification

In this section, the soundness verification experiments of our HDLM are evaluated, consisting of two types of experiments,

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(a) (b) (c)

Fig. 6. Feature Visualization under Limited Training Samples: Feature maps input to the classifier under different training samples each type of ELTS. Normal network means the normal network with SoftMax, our network means our network with dual loss. (a) Normal network Under 200 samples. (b) Normal network Under 20 samples. (c) Our network under 20 samples.

TABLE I
ORIGINAL IMAGE NUMBER OF DIFFERENT DEPRESSIONS FOR ELTS

Class	Training		Testing	
	Number	Depression	Number	Depression
BMP2-9563	233	17°	196	15°
BTR70-c71	233		196	
T72-132	232		196	
BTR60-7532	256		195	
2S1-b01	299		274	
BRDM2-E71	298		274	
D7-92	299		274	
T62-A51	299		273	
ZIL131-E12	299		274	
ZSU234-d08	299		274	

feature visualization under limited training samples and ablation experiments which typically refers to removing some innovations of the model and seeing the effects on the recognition performance of the proposed HDLM.

1) Feature Visualization Under Limited Training Samples: The training images were taken at a 17° depression angle, whereas the testing images were taken at a 15° depression angle. Table I summarizes the experimental setup for the training and testing datasets. The number of each target serial indicated in Table I is the number of original SAR images in the MSTAR dataset without the data augmentation. The training and testing images have the target at the center of the original images. Set the number of each target type as K, then the training images of each target are randomly selected from the original training dataset presented in Table I. Then, the K training images are augmented ten times by randomly sampling ten image chips of 88×88 SAR images from an original 128×128 SAR image to ensure the central target is complete [6]. These experiments are run under the ELTS conditions.

Fig. 6 shows the distributions of the features inputted into the classifier with the normal network under 200 and 20 samples each class. It is clear that though the normal network achieves the high recognition ratios 98.97% and obtain the discriminate feature distribution under 200 sample each class, the network fails to recognize precisely and discriminate in feature under 20 samples each class. At the same time, under 20 samples each class, our method still achieve high level of the discrimination in feature, which validates the effectiveness of our method in feature discrimination.

2) Ablation Experiments: The training details and settings are the same as these above. In the experiments of different training sample number, the training samples are randomly chosen from the original training dataset. We first run ablation experiments of our HDLM under different training samples, then we also presented the feature visualization of all the ablation configurations under 20 samples.

Table II shows the testing recognition ratios of different configurations of the proposed HDLM. The training epoch is set as 120 to ensure all the networks of different configurations converge. As shown in Table II, it is clear that the dual loss-based training looping and the architecture of the HDLM can improve the recognition performance in a wide range of the training sample number. Furthermore, the dual loss-based training looping can increase more recognition ratio to the architecture of the HDLM under the limited training samples, 5, 10, and 20 each class. At the same time, it seems that the dual loss and training looping can promote recognition ratio more obviously under insufficient dataset, 40 and 50 each class.

The recognition accuracy decreases when we remove some modules from the proposed architecture, and the decrease is particularly noticeable in experiments with only 20 and 10 training samples per class. Table II shows that the recognition ratios under 20, 10, and 5 of normal networks with dual loss are higher than the proposed structure without dual loss. It means that the dual loss can increase the recognition performance under the limited training samples.

Table II also show that the recognition ratios under 100 and 150 decreased to the level of normal CNNs, and the recognition ratios under 20, 10, and 5 seem like overfitting. It indicates that the proposed structure can achieve state-of-the-art with sufficient training samples and promote slight recognition performance with limited training samples.

Fig. 7 shows that the feature maps input to the classifier with different configurations. The training sample of each target under ELTS is 20. These feature maps are mapped to 2-D

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TABLE II
ABLATION EXPERIMENTS: TESTING RECOGNITION PERFORMANCE (%) OF ELTS

Methods	Labeled Number in Each Class							
	5	10	20	40	50	80	100	150
Ours	72.37	88.16	95.17	97.85	98.35	98.8	99.38	99.83
Our network with SoftMax	35.56	42.06	76.23	83.73	92.06	95.89	97.97	98.46
Normal network with DU loss	46.60	68.12	78.58	91.78	96.66	97.94	98.68	98.64
Normal network with SoftMax	30.68	37.75	67.22	81.64	88.29	93.47	96.97	99.05

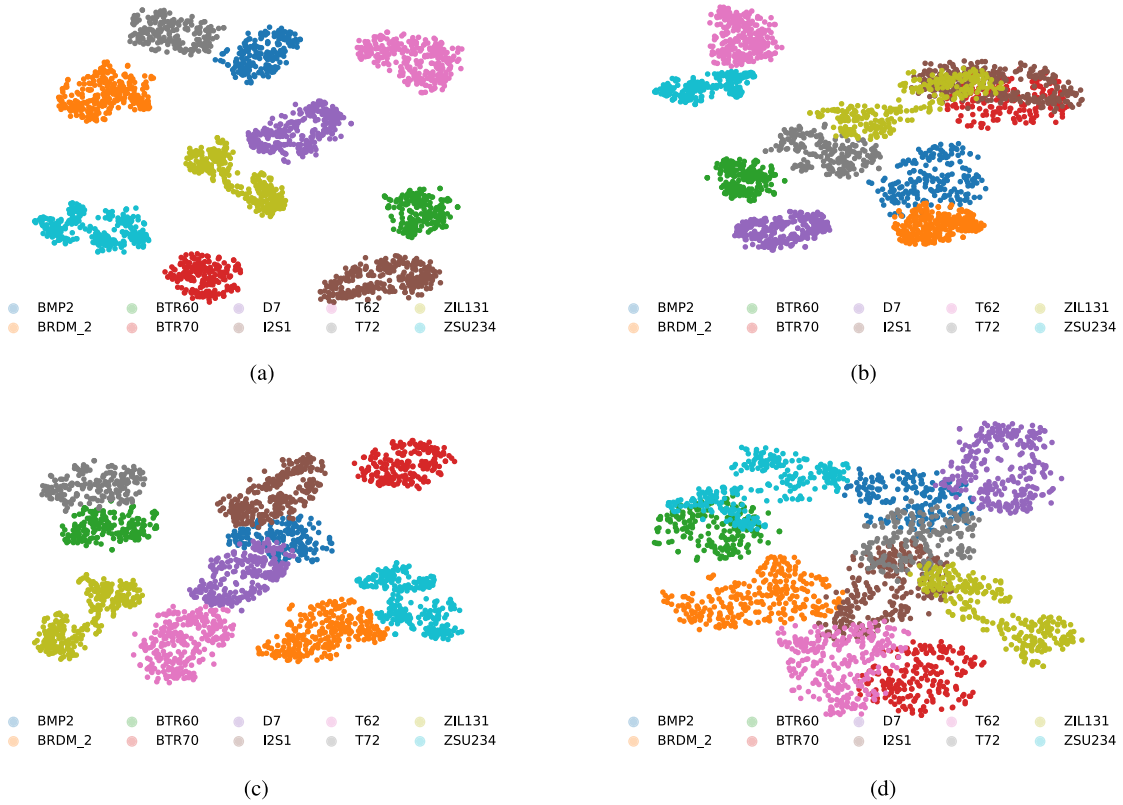


Fig. 7. Ablation experiments: Feature maps input to the classifier with different configurations under 20 each type of ELTS. (a) Ours. (b) Our network with SoftMax. (c) Normal network with dual loss. (d) Normal network with SoftMax.

TABLE III
IMAGE NUMBER OF DIFFERENT TARGETS IN OPENSARSHIP

Class	Training	Testing	Total
Bulk Carrier	300	374	674
Container Ship	300	710	1010
Tanks	300	253	553

space by t-distributed stochastic neighbor embedding (t-SNE). Fig. 7(a) and (b) show that the dual loss can increase the interclass distance and cluster the intraclass feature. From (a) and (c), the proposed network can acquire more interclass discrimination than intraclass clustering. From (b) and (d), the dual loss can obviously cluster the intraclass feature. Comparing (a)–(d), the interclass distance can be increased obviously when the network and dual loss work together. From these results above, it is clear that the proposed HDLM increases the recognition ratios under few training samples and promotes the discrimination of the extracted features.

C. Recognition Results of ELTS Under MSTAR

In the experimental setup of ELTS, the effectiveness of the proposed architecture is validated by the classification of ten targets in the MSTAR dataset. The training images and testing images were taken at a 17° depression angle and 15° depression angle. Table I summarizes the experimental setup for the training and testing datasets. The preprocess of the training and testing images is same as the ablation experiments.

Table IV presents the recognition result of the proposed HDLM for SAR ATR. The first row represents the number of training images in each target class. In the first column,

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TABLE IV
RECOGNITION PERFORMANCE (%) OF ELTS UNDER MSTAR

Class	Labeled Number in Each Class							
	5	10	20	40	50	80	100	150
BMP2-9563	68.21	87.18	95.90	99.49	98.97	97.43	98.46	99.48
BTR70-c71	67.52	85.77	94.89	98.54	97.45	98.17	100.00	100.00
T72-132	56.41	84.62	94.87	97.44	99.49	97.43	97.43	99.48
BTR60-7532	54.59	89.80	94.90	96.94	97.96	100.00	100.00	100.00
2S1-b01	75.91	92.34	97.81	99.27	98.91	99.63	98.9	99.63
BRDM2-E71	84.67	93.80	98.91	99.64	95.99	97.44	99.63	100.00
D7-92	86.81	92.31	95.60	99.27	97.07	98.16	98.9	99.63
T62-A51	69.90	86.73	99.49	97.96	98.98	99.48	100.00	100.00
ZIL131-E12	73.72	87.59	89.42	90.15	99.64	100.00	100.00	100.00
ZSU234-d08	74.45	80.29	91.24	100.00	99.64	100.00	100.00	100.00
Average	72.37	88.16	95.17	97.85	98.35	98.8	99.38	99.83

the class and series of the target are linked with hyphens. All the recognition ratios of the ten targets in MSTAR and the average recognition ratio are presented in Table IV. When the training sample of each class in MSTAR is 150, 100, and 80, respectively, the recognition rates are 99.82%, 99.38%, and 98.80%. It means that when the training sample is enough to train the model, the proposed HDLM for SAR ATR achieves state-of-the-art. When the training sample of each class in MSTAR is 50, 40, and 20, the recognition rates are 98.25%, 97.81%, and 95.17%, respectively. It means that when the training samples decrease gradually, the recognition rates are still stable and robust for the few training samples. When the training sample of each class in MSTAR is 10 and 5, the recognition rates are 88.16% and 72.37%, respectively. When the number of training samples is too few, the proposed HDLM can still achieve a good recognition rate. Such a situation is often what the actual situation will face, which is the key difficulty of practical application.

Moreover, Table V compares ELTS under MSTAR with other methods to better demonstrate the effectiveness of the proposed method. For recognition, the SAR ATR algorithms used for comparison are described as follows. PCA + support vector machine (SVM), label consistent (LC)-KSVD, discriminative graphical model (DGM), and AdaBoost [33] are traditional algorithms. The two deep neural networks (DNNs) [34], [35] utilize classical DNNs and achieve the state-of-the-art performance in SAR ATR. CNN1 [33] and CNN2 [36] are normal frameworks for classification, while CNN1 combines the generative adversarial network (GAN) with CNN. CNN + matrix [36] combines the CNN and matrix to achieve high performance, while GAN-CNN [33] and multi-discriminator GAN (MGAN)-CNN [33] employ GAN to generate fake SAR images and achieve high recognition performance. Semisupervised learning [20] utilizes unlabeled data during training to alleviate the demand for labeled data.

TABLE V
COMPARISON OF PERFORMANCE (%) OF ELTS UNDER MSTAR

Method	Labeled Number in Each Class			
	All	80	40	20
PCA+SVM [33]	94.32	92.48	87.95	76.43
ADaboost [33]	93.51	91.45	86.45	75.68
LC-KSVD [33]	95.13	93.23	87.39	78.83
DGM [33]	96.07	92.85	88.14	81.11
DNN1 [34]	95.54	93.04	86.98	77.86
DNN2 [35]	96.50	93.76	87.73	79.39
CNN1 [33]	97.03	93.88	88.35	81.80
CNN2 [36]	-	-	-	75.88
CNN+matrix [36]	-	-	-	82.29
GAN-CNN [33]	97.53	94.91	90.13	84.39
MGAN-CNN [33]	97.81	94.91	90.82	85.23
Semisupervised Learning [20]	-	98.65	97.11	92.62
The Proposed Method	-	**98.80**	**97.85**	**95.17**

The quantitative experiment shows that the proposed methods achieve higher recognition rates than other methods under any number of training images in each class. In conclusion, the proposed HDLM can achieve the state-of-the-art recognition performance from sufficient training samples to few training samples.

The above results and analysis show that the proposed model can achieve state-of-the-art performance with sufficient training samples and acquire high recognition rates when facing few training samples, even the five-shot or ten-shot. This can promote the practical applications of SAR ATR.

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TABLE VI
RECOGNITION PERFORMANCE (%) OF ELTS UNDER OPENSARSHIP

Class	Labeled Number in Each Class					
	10	50	60	75	100	150
Bulk Carrier	51.60	59.09	59.63	62.83	66.04	74.60
Container Ship	81.69	83.52	86.76	82.25	90.42	89.72
Tanks	51.78	86.17	77.08	88.93	92.09	88.93
Average	69.00	78.78	78.93	79.69	85.65	87.01

D. Recognition Results of ELTS Under OpenSARShip

Following [20], [36], the three-class objects, bulk carrier, container ship, and tanks, are chosen from OpenSARShip, as shown in Table III. The amounts of three-class objects are imbalanced. To prevent the influence of class imbalance, we set the number of training samples equal for each class, according to the least number of samples in all three categories, with the training–testing ratio as 4:6. Moreover, the remaining samples are regarded as testing samples. The training image is adjusted to a 128 $\times$ 128 SAR image and then augmented for ten times by randomly sampling ten image chips of 88 $\times$ 88 SAR images from an original 128 $\times$ 128 SAR image [6].

Table VI presents the recognition result of the proposed HDLM. These results show that when training samples are sufficient, the recognition of the proposed HDLM achieves the state-of-the-art, like 87.01% under 150 each target and 85.65% under 100 each target. Table VII shows the comparisons with other methods. The algorithms used for comparison in OpenSARShip are the same as that used in MSTAR, and the quantitative results are obtained from [20] and [36]. According to the number of training images in each class, we delineate three bands (from 1 to 50, from 51 to 100, and from 101 to 240). When the recognition ratios of other methods are achieved above 85%, the training samples are much larger than ours. When the training samples are few, the proposed HDLM achieve 69.00% under 10 each target, 78.78% under 50 each target, which achieved the best performance compared with other methods. Even for the recognition performance of our method in the second band (51 100), it is remarkably better than that of other methods in the third band (101 240).

The results of ELTS under the MSTAR and OpenSARship dataset validate the robustness and effectiveness of the proposed HDLM. It is clear that under the experiments of sufficient training samples and few training samples, the proposed HDLM is effective and robust in achieving the state-of-the-art recognition performance facing different SAR datasets. Then, the effectiveness of EFSL is validated as follows.

E. Recognition Results of EFSL Under MSTAR

We conducted experiments on the large depression angle difference between training and testing samples under FSL of five-shot training data. The experiment contains three target classes in MSTAR for testing (2S1, BRDM-2, and zenitnaya samokhodnaya ustanovka (ZSU)-23/4) acquired at a large depression angle of 30°, meaning larger differences in SAR images. For five-shot training samples, the three target types are acquired at a depression angle of 17°. The training image is adjusted to a 128 $\times$ 128 SAR image and then augmented for ten times through randomly sampling ten image chips of 88 $\times$ 88 SAR images from an original 128 $\times$ 128 SAR image [6]. Table VIII shows the distribution of testing data in this experiment [37]. Table IX shows the recognition confusion matrices.

TABLE VII
COMPARISON OF PERFORMANCE (%) OF ELTS UNDER OPENSARSHIP (THE NUMBER IN PARENTHESES IS THE NUMBER OF THE TRAINING SAMPLES FOR EACH METHOD)

Methods	Number range of training images in each class		
	1 to 50	51 to 100	101 to 240
Semi-Supervised [20]	61.88 (20) 64.73 (40)	68.67 (80)	71.29 (120) 74.96 (240)
Supervised [20]	58.24 (20) 62.09 (40)	65.63 (80)	68.75 (120) 70.83 (240)
CNN [36]	62.75 (50)	68.52 (100)	73.68 (200)
CNN+Matrix [36]	72.86 (50)	75.31 (100)	77.22 (200)
Proposed	**69.00 (10)** **78.78 (50)**	**78.93 (60)** **76.69 (75)** **85.65 (100)**	**87.01 (150)**

TABLE VIII
TESTING NUMBER FOR DIFFERENT CLASSES IN ABLATION EXPERIMENT

Depression Angle	Target Class	Testing Number
15°	2S1	274
	BRDM2	274
	ZSU234	274
30°	2S1	288
	BRDM2	257
	ZSU234	287

The results of ELTS and EFSL show that the proposed method can obtain state-of-the-art recognition results with a limited training sample dataset. Moreover, with the limited training samples for all classes or FSL settings, the information from the training data can be fully utilized to optimize the recognition of FSL. Therefore, the proposed method has the robustness to handle the complex practical application of SAR ATR methods in FSL and can confront a variety of the limited training samples to enhance the feasibility and practicability of recognition methods in practical applications.

In conclusion, through the ablation experiments, the effectiveness of our method has been validated under limited

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TABLE IX
CONFUSION MATRIX OF EFSL UNDER MSTAR

5 shot	Test on 15°			Test on 30°		
	2S1	BRDM2	ZSU234	2S1	BRDM2	ZSU234
2S1	**91.97**	2.55	5.47	**89.24**	7.64	3.12
BRDM2	3.28	**91.61**	5.10	13.61	**84.43**	1.94
ZSU234	0	0.36	**99.64**	12.5	0.69	**86.80**

training data. Furthermore, facing different SAR datasets, the proposed HDLM can achieve the state-of-the-art performance with sufficient training samples and acquire high recognition rates when facing various situations with the limited training samples.

IV. CONCLUSION

The practical application of SAR ATR methods faces various distributions of the training samples for each target. The novel HDLM is proposed to tackle the problems of limited samples per class in SAR ATR. The experimental results on the MSTAR and OpenSARship datasets of EFSL and ELTS show that the proposed HDLM significantly improves the recognition performance. From the ablation study, the dual loss and the structure of HDLM have different effects on the recognition of the limited training sample. The dual loss can cluster the extracted feature in the same class and help the proposed HDLM to exploit the mutual information in the same class. The structure of the HDLM can extract the distinguishable feature among different classes and enhance the intermediate representation learning with natural feature discrimination. Each part of the HDLM helps each other to achieve the optimal recognition performance and tackle the problems of the limited data in SAR ATR.

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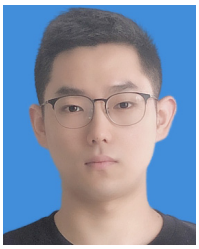
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